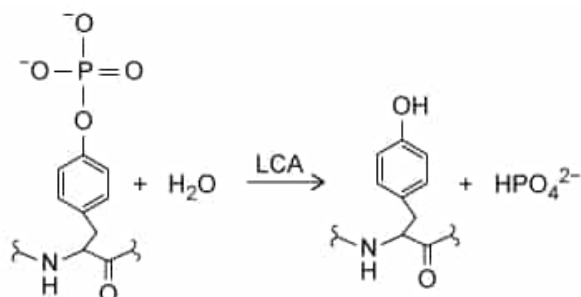


Leukocyte common antigen (LCA) enzymes play several important developmental roles in both vertebrates and invertebrates. They regulate numerous cellular processes by catalyzing Reaction 1 in target proteins at specific developmental stages. Consequently, LCA enzyme expression and activity are tightly controlled in various cell types.



Reaction 1

The Src protein is critical for cell differentiation and motility, and is activated by LCA enzyme activity. Two LCA enzymes, known as LAR and CD45, were purified and incubated with the phosphopeptide src527 (Figure 1), derived from Src.

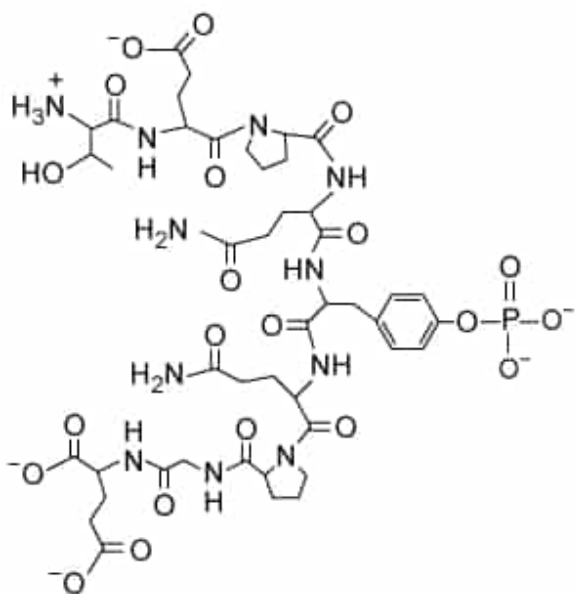


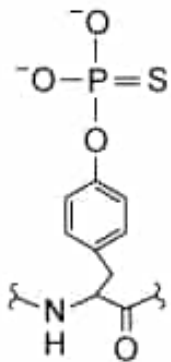
Figure 1 Structure of the src527 peptide used as a substrate in LAR and CD45 assays

The rate of Reaction 1 was measured at various concentrations of src527 in the presence of either LAR or CD45, and kinetic parameters were determined (Table 1).

Table 1 Kinetic Parameters of LAR and CD45

Enzyme	[E] (μM)	V_{max} ($\mu\text{M/s}$)	K_{M} (μM)	Catalytic efficiency ($\text{s}^{-1} \mu\text{M}^{-1}$)
LAR	0.050	2.0	353	0.11
CD45	0.010	2.8	162	1.73

Compound **1** was synthesized and tested for its effectiveness as an inhibitor of LCA enzymes.



Compound 1

The study in Table 1 was repeated in the presence of Compound 1, which was found to significantly reduce the activities of both LAR and CD45 against the src527 peptide. Compound 1 was subsequently used in assays to examine the *in vivo* function of LCA enzymes.

Which of the following calculations was used to determine the catalytic efficiencies in Table 1?

- ☐ A. Multiplying $V_{\max}$ by K_m [1%]
- ☐ B. Multiplying K_m by $V_{\max}$, then dividing the result by $[E]$ [9%]
- ☒ C. Dividing $V_{\max}$ by $[E]$, then dividing the result by K_m [82%]
- ☐ D. Dividing K_m by $V_{\max}$ [6%]

Correct

83% answered correctly

Explanation:

Calculating catalytic efficiency

$$\frac{V_{\max}}{[E]} = k_{\text{cat}} \rightarrow \frac{k_{\text{cat}}}{K_M} = \text{catalytic efficiency}$$

Enzyme	[E] (μM)	$V_{\max}$ ($\mu\text{M/s}$)	K_M (μM)	Catalytic efficiency ($\text{s}^{-1} \mu\text{M}^{-1}$)
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Enzyme activity is characterized by several parameters, including k_{cat} , K_m , and **catalytic efficiency**. k_{cat} is equal to $V_{\max}$ divided by the total enzyme concentration $[E]$, and is proportional to the enzyme rate at high substrate concentrations. K_m is the substrate concentration at which $\frac{1}{2}V_{\max}$ occurs. **Catalytic efficiency** is the **ratio k_{cat}/K_m** and is a measure of an enzyme's **specificity for a substrate**. This ratio is proportional to the reaction rate at low substrate concentrations and is often used to compare two enzymes that catalyze the same reaction.

Table 1 gives the K_m values of LAR and CD45 but does not give k_{cat} . However, the table does give the maximum reaction velocity V_{max} and the total enzyme concentration $[E]$. The value of k_{cat} is equal to $V_{max}/[E]$; therefore, k_{cat} can be calculated by dividing the given V_{max} value by the given $[E]$. Catalytic efficiency can then be determined by dividing the resulting k_{cat} value by K_m .

(Choices A and B) *Multiplying K_m by V_{max} is not a step in the calculation of catalytic efficiency. K_m is the denominator of catalytic efficiency, k_{cat}/K_m .*

(Choice D) The K_m/V_{max} ratio is shown in the slope of [Lineweaver-Burk](#) plots and is proportional to the *inverse* of catalytic efficiency.

Educational objective:

Catalytic efficiency measures the specificity of an enzyme for a substrate and is calculated as k_{cat}/K_m . k_{cat} can be calculated by dividing V_{max} by the total enzyme concentration $[E]$.

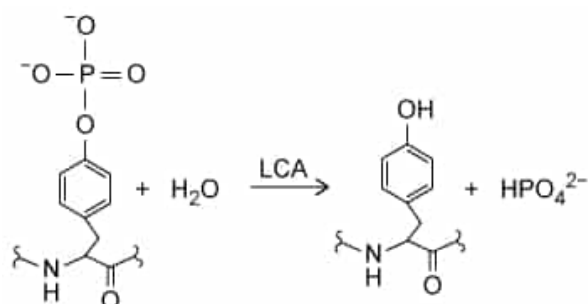
Time spent: 0 sec

Subject:
Biochemistry

QID: 402171

System:
Enzymes

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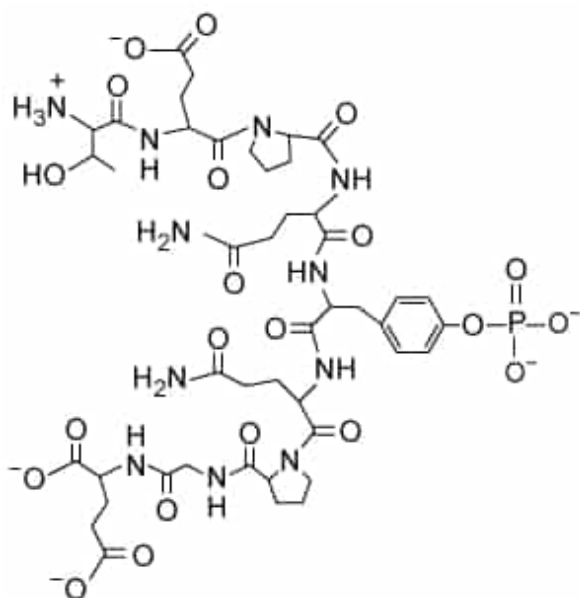


Figure 1 Structure of the src527 peptide used as a substrate in LAR and CD45 assays

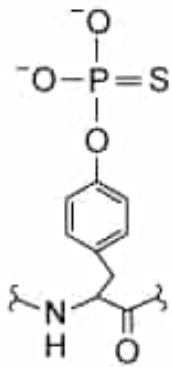
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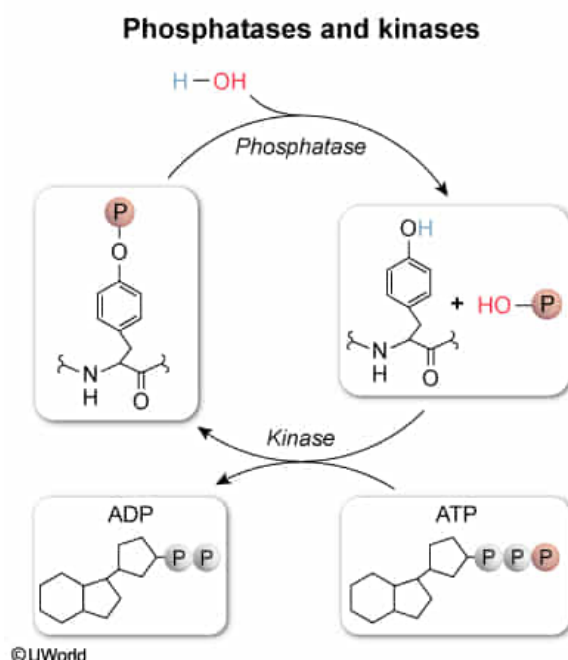
The information in the passage suggests that the effects of LCA activity could be counteracted by which type of enzymatic activity?

- ☐ A. Dehydrogenase activity [14%]
- ✓ ☒ B. Kinase activity [73%]
- ☐ C. Synthetase activity [5%]
- ☐ D. Mutase activity [6%]

Correct

73% answered correctly

Explanation:



Enzymes are divided into **classes and subclasses** based on the kinds of reactions they catalyze. Biological processes are frequently regulated by opposing reactions such as phosphorylation and dephosphorylation, with each reaction counteracting the other. **Phosphorylation reactions** are carried out most commonly by **kinases**, a type of **transferase** that moves a phosphate group (HPO_4^{2-}) from a **donor molecule** (usually ATP) to an acceptor molecule. **Dephosphorylation reactions** are catalyzed by **phosphatases** that remove an inorganic phosphate group by adding water across a bond (hydrolysis).

Reaction 1 shows that LCA enzymes catalyze the **hydrolysis** of the phosphoester bond in phosphotyrosine residues. Enzymes that hydrolyze bonds are known as **hydrolases**, and the subset of **hydrolases that remove phosphate groups** (including LCA enzymes) are classified as phosphatases. The question asks which type of enzymatic activity most likely counteracts the effects of LCA enzymes. Because LCA enzymes are phosphatases that facilitate the removal of phosphate groups, their activity is most likely **counteracted by kinase activity, which adds phosphate**.

(Choice A) Dehydrogenases are a type of **oxidoreductase** that transfer electrons through hydrogen atoms or hydride ions. They do not counteract phosphatase activity.

(Choice C) Synthetases are a type of **ligase** that link two molecules together, and typically use ATP hydrolysis as a source of energy. They should not be confused with synthases, which are **lyases** and do not require energy input. Neither synthetases nor synthases phosphorylate acceptor molecules.

(Choice D) Mutases are a type of **isomerase** that rearrange functional groups within a molecule. They do not counteract phosphatase activity.

Educational objective:

Phosphatases catalyze the dephosphorylation of their substrates by hydrolysis, producing inorganic phosphate. Kinases catalyze the phosphorylation of their substrates by transferring organic phosphate from a donor molecule, usually ATP. The two enzyme types often regulate biological processes by counteracting each others' effects.

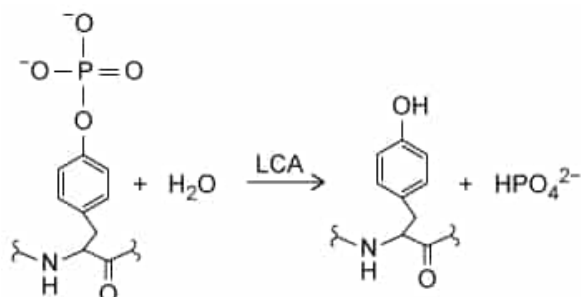
Time spent:0 sec

Subject:
Biochemistry

QID:402172

System:
Enzymes

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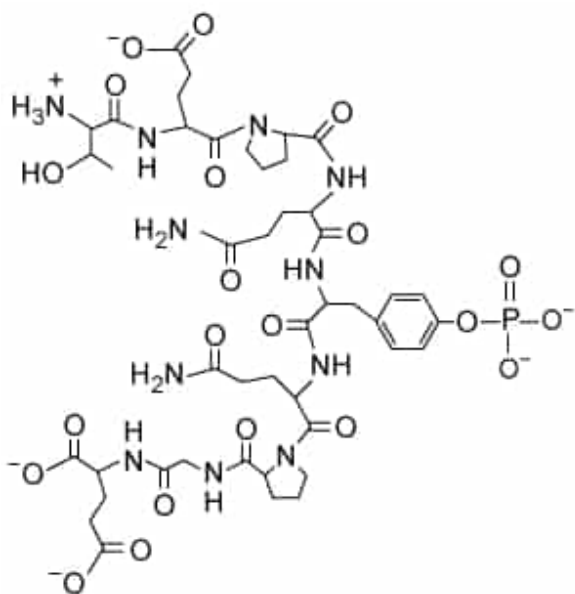


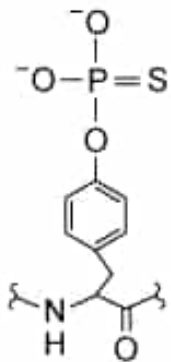
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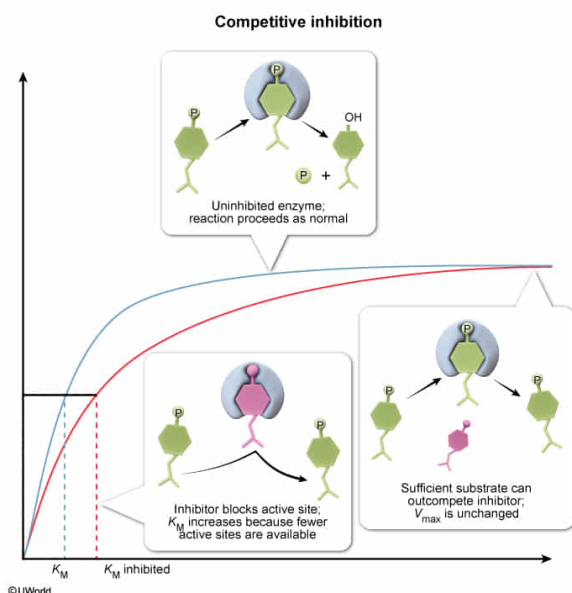
Compound 1 would most likely have what effect on the activity of LAR?

- ☐ A. It would bind to an allosteric site and decrease the K_m . [11%]
- ☐ B. It would bind to the active site and decrease the V_{max} . [22%]
- ☐ C. It would bind to an allosteric site and increase the V_{max} . [3%]
- ☒ D. It would bind to the active site and increase the K_m . [62%]

Correct

62% answered correctly

Explanation:



Inhibitors can decrease the activity of enzymes in a **variety of ways**, depending on their mechanism of action. **Competitive inhibitors** work by **binding to the active site** of an

enzyme, resulting in the exclusion of the substrate from the active site. Because

enzyme active sites are typically specific to one molecule or a few similar molecules, competitive inhibitors usually **mimic the substrate's structure**.

A comparison of Compound **1** and the substrate shown in Reaction 1 reveals that they have **similar structures**. The only difference between the two molecules is the presence of a sulfur atom in Compound **1** in place of oxygen. Therefore, Compound **1** most likely is **similar in shape to the substrate**, allowing Compound **1** to bind to the same site as the substrate (ie, the active site) and **act as a competitive inhibitor**. These types of inhibitors **increase the amount of substrate needed** to achieve $V_{\max}$, as well as the amount of substrate needed to achieve $\frac{1}{2}V_{\max}$.

K_m is defined as the concentration of substrate needed to reach $\frac{1}{2}V_{\max}$, so **competitive inhibitors increase K_m** . However, if the concentration of substrate is high enough, it can outcompete a competitive inhibitor and $V_{\max}$ can be achieved, so **competitive inhibitors do not alter $V_{\max}$** . Therefore, Compound **1** is expected to bind LAR at the active site and increase the K_m .

(Choice A) Competitive inhibitors *increase* K_m and bind at the *active* site rather than at an allosteric site.

(Choice B) Competitive inhibitors do not alter $V_{\max}$. Instead, they increase the amount of substrate required to achieve $V_{\max}$, yielding an increased K_m .

(Choice C) A molecule that binds at an allosteric site and increases $V_{\max}$ is an activator, not an inhibitor. The passage identifies Compound **1** as an inhibitor.

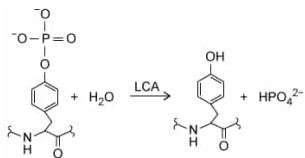
Educational objective:

Competitive inhibitors typically bind enzymes at the active site, and therefore are usually similar in structure to the substrate. They have no effect on $V_{\max}$ but increase the amount of substrate needed to achieve $V_{\max}$, resulting in an increased K_m value.

Time spent:0 sec
Subject:
Biochemistry

QID:402173
System:
Enzymes

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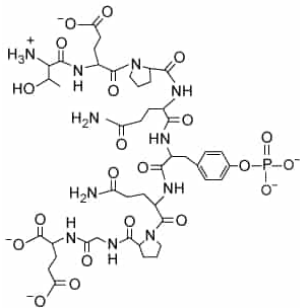


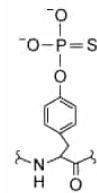
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Based on Table 1, the turnover number (*k*_{cat}) of CD45 exceeds that of LAR by what factor?

- ☐ A. 1.5 [15%]
☐ B. 2 [13%]
☐ C. 5 [13%]
☒ D. 7 [57%]

Correct
56% answered correctly

Explanation:

Calculating turnover number (*k*_{cat})

Enzyme	[E] (μM)	V _{max} (μM/s)	<i>k</i> _{cat} (s ⁻¹) = V _{max} (μM/s) / [E] (μM)
LAR	0.050	2.0	40 (2.0/0.05)
CD45	0.010	2.8	280 (2.8/0.01)

Turnover number ratio: 280/40 = 7

The **turnover number** *k*_{cat} of an enzymatic reaction is the number of **reactions catalyzed per second per enzyme** under **saturating conditions** (ie, when *V*_{max} is achieved). It is useful for comparing enzymes when substrate concentration is high, and can be calculated by **dividing *V*_{max} by the total enzyme concentration [E]**.

Table 1 does not show the turnover numbers for either CD45 or LAR, but each can be calculated from the given *V*_{max} and [E]:

- For CD45, *V*_{max} = 2.8 μM/s and [E] = 0.01 μM. Therefore, *k*_{cat} = $\frac{2.8 \text{ μM/s}}{0.01 \text{ μM}} = 280 \text{ s}^{-1}$.

- For LAR, $V_{\max} = 2.0 \mu\text{M/s}$ and $[E] = 0.05 \mu\text{M}$. Therefore, $k_{\text{cat}} = \frac{2.0 \mu\text{M/s}}{0.05 \mu\text{M}} = 40 \text{ s}^{-1}$.

The ratio of the two k_{cat} values is $\frac{280}{40} = 7$. Therefore, the turnover number of CD45 exceeds that of LAR by a **factor of 7**.

(Choice A) The ratio of $V_{\max}$ of CD45 to $V_{\max}$ of LAR is approximately 3:2, so $V_{\max}$ of CD45 is approximately 1.5 times greater than that of LAR. However, CD45 is also present at a concentration $[E]$ that is 5 times smaller than that of LAR. Because $k_{\text{cat}} = V_{\max}/[E]$, the enzyme concentrations must be considered.

(Choice B) The ratio of LAR K_m to CD45 K_m is approximately 2:1, but this is not the ratio of turnover numbers (k_{cat}).

(Choice C) The concentration of LAR exceeds that of CD45 by a factor of 5, but enzyme concentration $[E]$ is not the turnover number.

Educational objective:

The turnover number k_{cat} of an enzyme-substrate system represents the number of reactions catalyzed per second per enzyme under saturating conditions. It is calculated as the $V_{\max}$ divided by the total enzyme concentration $[E]$. k_{cat} can be used to compare enzymes at high substrate concentrations.

Time spent: 0 sec

Subject:
Biochemistry

QID: 402174

System:
Enzymes